

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

PAPER 5 Practical Test

9189/5

NOVEMBER 2017 SESSION

1 hour 20 minutes

Candidates answer on the question paper.
Additional materials:
as listed in Instructions to Supervisors.

TIME 1 hour 20 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
You are advised to show all working in calculations.
Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 8 printed pages.

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- 1 You are required to determine the enthalpy change of decomposition of NaHCO_3 by carrying out experiments 1, 2 and 3.

FA1 is $3.00 \text{ mol dm}^{-3} \text{ HCl}$

FA2 is a mixture of $1 \text{ g Na}_2\text{CO}_3$ and 4 g NaHCO_3

FA3 is a mixture of $2.5 \text{ g Na}_2\text{CO}_3$ and 2.5 g NaHCO_3

FA4 is a mixture of $4 \text{ g Na}_2\text{CO}_3$ and 1 g NaHCO_3

(a) (i) **Experiment 1:**

Fill the burette with **FA1**.

Place a plastic cup in a 400 cm^3 beaker.

Add 35.00 cm^3 of **FA1** into the plastic cup.

Measure and record, in **Table 1.1** the initial temperature of **FA1**.

Transfer all **FA2** into the plastic cup.

Stir with the thermometer.

Measure and record, in **Table 1.1** the highest/lowest temperature reached.

Table 1.1

final temperature/ $^{\circ}\text{C}$	
initial temperature/ $^{\circ}\text{C}$	
change in temperature/ $^{\circ}\text{C}$	

[3]

(ii) **Experiment 2:**

Repeat experiment 1 using **FA1** and **FA3**.

Record the results in **Table 1.2**.

Table 1.2

final temperature/ $^{\circ}\text{C}$	
initial temperature/ $^{\circ}\text{C}$	
change in temperature/ $^{\circ}\text{C}$	

[3]

(iii) Experiment 3

Repeat experiment 1 using **FA1** and **FA4**.

Record the results in **Table 1.3**.

Table 1.3

final temperature/°C	
initial temperature/°C	
change in temperature/°C	

[3]

- (b) (i)** Complete **Table 1.4** by showing the percentage (%) by mass of Na_2CO_3 used in each experiment and the corresponding change in temperature, ΔT .

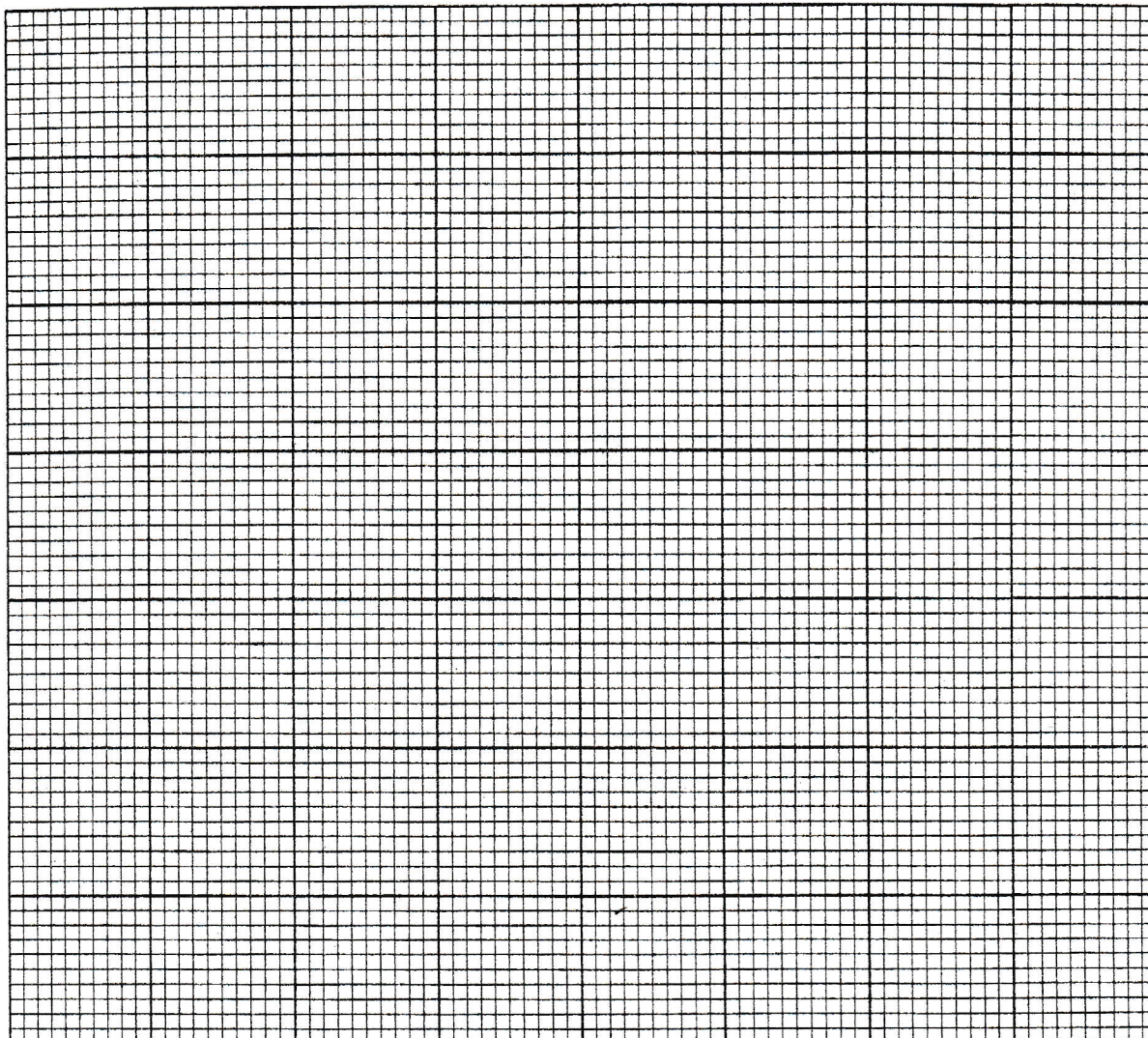
Table 1.4

Experiment number	% by mass of Na_2CO_3 in mixture	change in temperature, ΔT /°C
1		
2		
3		

[3]

- (ii) Draw the graph of change in temperature, ΔT , against the % by mass of Na_2CO_3 used.

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[6]

- (iii) Describe the relationship between % by mass of Na_2CO_3 and temperature change.

_____ [1]

- (iv) Determine the temperature change, ΔT , when

1. Na_2CO_3 is used on its own,

_____ [1]

2. NaHCO_3 is used on its own.

_____ [1]

- (v) Use the values in (b)(iv) to calculate the molar enthalpy change for the reaction between HCl and

1. Na_2CO_3 ,

[3]

2. NaHCO_3 .

[3]

- (c) (i) Construct an energy cycle involving the reactions in (b)(v) to obtain the enthalpy change for the thermal decomposition of $\text{NaHCO}_{3(s)}$ to form Na_2CO_3 , H_2O and CO_2 .

[2]

- (ii) Hence determine the molar enthalpy change for the thermal decomposition of $\text{NaHCO}_{3(s)}$.

[1]

[Total: 27]

ASSESSMENT OF DESIGN AND PLANNING SKILLS

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DO NOT CARRY OUT YOUR PLAN

- 2 A fungicide for plant diseases consists of a mixture of copper (II) sulphate and calcium hydroxide.

You are required to devise a procedure that can be used to determine the amount of copper (II) sulphate present in a sample of the fungicide.

Assuming you are provided with the following and any other laboratory equipment of your choice present your plan as a sequence of numbered step.

5 g sample of the fungicide
0.1 mol dm⁻³ NaOH
2.0 mol dm⁻³ HCl
100 cm³ measuring cylinder
250 cm³ volumetric flask
250 cm³ conical flask

The Plan